The empirical recurrence for OEIS A296112: recovering a conjecture that is not written down

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Abstract

OEIS A296112 records "Empirical recurrence of order 89", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on $S = 1056$ vertices and satisfies some monic recurrence of order at most S; Berlekamp–Massey applied to $2S$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 89, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 89 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A296112 is "Number of n X 5 0..1 arrays with each 1 adjacent to 3 or 4 king-move neighboring 1's.". It has offset 1 and begins

$$1, 8, 26, 64, 503, 2800, 14057, 106920, \ \ldots$$

The entry states:

> Empirical recurrence of order 89 (see link above).

As of the "Last modified" line on the live entry (Jul 24 2024 17:33:59, revision 10) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 1056$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S = 1056$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 2112$ exact terms of the model returns order 89 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{89} c_i\, a(n-i),$$

c_{1}	3	c_{24}	1396971345	c_{47}	1198222490579	c_{70}	−117595727125
c_{2}	13	c_{25}	8251033161	c_{48}	2648177984692	c_{71}	140806178830
c_{3}	201	c_{26}	2593006103	c_{49}	3499154722608	c_{72}	−7066306343
c_{4}	−229	c_{27}	8719513887	c_{50}	2416243861280	c_{73}	−31664807646
c_{5}	−1207	c_{28}	−12515507918	c_{51}	1014638080035	c_{74}	−16041256920
c_{6}	−13279	c_{29}	1367773474	c_{52}	−1637425495650	c_{75}	6316659156
c_{7}	−1974	c_{30}	−36148446183	c_{53}	−2087951675530	c_{76}	8270720208
c_{8}	13528	c_{31}	8366199970	c_{54}	−2467102465530	c_{77}	−378016789
c_{9}	351028	c_{32}	−74199853620	c_{55}	−1143420016985	c_{78}	−2121221148
c_{10}	299293	c_{33}	33328326278	c_{56}	−1212987906421	c_{79}	1785870210
c_{11}	628008	c_{34}	−115941616725	c_{57}	−701748686658	c_{80}	75657602
c_{12}	−4034183	c_{35}	161615515153	c_{58}	−436867671924	c_{81}	69250066
c_{13}	−5808716	c_{36}	49880561129	c_{59}	962641855937	c_{82}	−104622294
c_{14}	−17078917	c_{37}	560617936488	c_{60}	1941161938176	c_{83}	−36270612
c_{15}	15358166	c_{38}	261357731043	c_{61}	575158077199	c_{84}	6610992
c_{16}	39594712	c_{39}	833930434903	c_{62}	−1025596414109	c_{85}	4314788
c_{17}	168873366	c_{40}	−182684724911	c_{63}	535645822767	c_{86}	800696
c_{18}	81312802	c_{41}	445456225227	c_{64}	386297035089	c_{87}	−764384
c_{19}	47888202	c_{42}	−764207217858	c_{65}	129658974306	c_{88}	54144
c_{20}	−811249396	c_{43}	−147922088521	c_{66}	−779740466976	c_{89}	640
c_{21}	−865549753	c_{44}	−1657486777160	c_{67}	389547401591		
c_{22}	−1777602907	c_{45}	−756365070131	c_{68}	−183266014526		
c_{23}	1032623714	c_{46}	−1325706151921	c_{69}	−9759787372		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 89, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $89 + 4$ terms removed returns order 89 as well, so $\deg q_{\min} = 89 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 24 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $89+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 89 that this sequence satisfies — and that family turns out to have exactly one member.

References

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